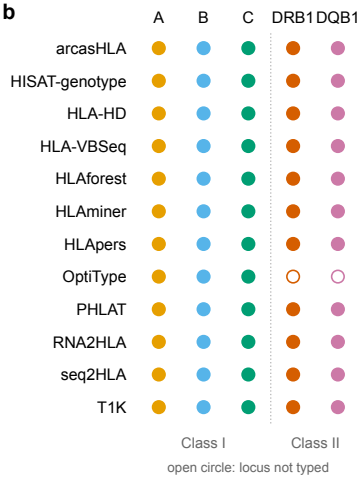
a

Dataset	n	Cells	Loci	Reads (bp)	Gold standard
D1	490	LCL	A B C DRB1 DQB1	2 x 75	PCR-SBT
D2	86	PBMC and sorted subsets	DRB1	2 x 126	PCR-SSP
D3	50	LCL	A B C	2 x 37	PCR-SSOP
D4	14	Naive CD4+ T cells	DRB1	2 x 126	PCR-SSP
D5	8	B721.221, one allele	C	2 x 38	Single-allele transduction
D6	4	B721.221, one allele	A or B	2 x 37	Single-allele transduction
D7*	20	Bone marrow (BMMC)	A B C DRB1 DQB1	10x 3'	NGS amplicon (Scisco)
D8*	3	Whole blood	A B C DRB1 DQB1	2 x 151	PacBio HiFi assembly

675 samples. * outside the D1-D6 main scope (partial caller coverage or n = 3). D5-D6: HLA-null B721.221 lines each carrying one transduced allele.

b



c

correct miscalled no call

Gold standard (D1 sample ERR188023, HLA-B)

`B*13:02  B*56:01/B*56:24`

an entry with alternatives ("/") matches any of them

	Call 1	Call 2	Outcome (2-field names compared)	Accuracy
T1K	B*13:02:01 -> B*13	B*56:01:01 -> B*56:02	correct	**2 / 2**
seq2HLA	B*55:02	B*13:02	1 correct, 1 miscalled	**1 / 2**
PHLAT	B*13:03	B*55:50	2 miscalled	**0 / 2**
HLAminer	no call	no call	2 no call	**0 / 2**

Accuracy = correct calls / all calls (two per sample and locus; no-calls count in the denominator). Calls are truncated to 2 fields (protein) or 1 field (allele group) before names are compared; each gold-standard allele can be matched by one call only.